Woodturning

National 5 · Machines, materials & finishing · Worksheet

WOODWORK WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ prepare and mount a blank for between-centres turning
- ☐ explain faceplate and between-centres turning
- ☐ use turning tools and check diameters with calipers
- ☐ use the lathe safely

I DO — Worked example

Read how a full-mark answer is built

A pupil prepares a square blank for between-centres turning.

1 Explain why diagonal lines are drawn and sawn on one end of the blank. (2 marks)

Full-mark answer each ✓ is one mark

Where the diagonals cross is the centre of the blank, so it is mounted centrally ✓. The saw cuts let the prongs of the drive centre grip the wood so it doesn't slip when turning ✓.

Why marks are not awarded

"To find the middle."

1 mark: explains the lines but not the saw cuts.

WE DO — Guided practice

Use the starters and word bank to help you

A bowl is to be turned on the lathe.

2 Name the method of turning used for a bowl. (1 mark)

Sentence starters	Word bank
1. …	Faceplate turning

Tip: 1 marks = 1 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) How would you check a turned diameter?

☐ Outside calipers ☐ Try square ☐ Marking gauge ☐ Sliding bevel

b) Which method would you use to turn a bowl?

☐ Between centres ☐ Faceplate ☐ Pillar drill ☐ Mortiser

c) What should you always do before switching on the lathe?

☐ Sand the wood all over with fine abrasive paper ☐ Apply the first coat of finish to the blank ☐ Remove the tailstock so there's more room ☐ Rotate the work by hand to check it clears the tool rest

d) Why are saw cuts made along the diagonals on one end of a blank?
☐ So the drive centre grips and doesn't slip ☐ To decorate the end of the blank ☐ To make the blank lighter to turn ☐ To mark the length of the finished part

4 Key terms: write the letter of the matching description next to each term.

1. **Between-centres turning**	____	**A** Find the centre of each end
2. **Faceplate turning**	____	**B** Wood held between drive centre and tailstock centre: legs, handles
3. **Diagonal lines on the blank**	____	**C** Wood screwed to a faceplate: bowls, bases
4. **Sawing along the diagonals on one end**	____	**D** Turns a square blank round
5. **Roughing gouge**	____	**E** So the drive centre's prongs grip and the wood doesn't slip
6. **Parting tool**	____	**F** Cuts grooves to a diameter and parts work off

A pupil is about to turn a table leg.

5 Describe two safety precautions when using the woodturning lathe. (2 marks)

A turned handle must be 25 mm in diameter.

6 Describe how the diameter is checked during turning. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?